

Bengali Word Assimilation Integrating Using Deep Learning Methods

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Abstract— Natural language processing makes heavy use of topic modeling approaches to derive subjects from unstructured text input. However, Bangla's NLP (natural language processing) technology is not as advanced as English's, and not much research has been done about the Bengali dialect—one of the most widely spoken languages in the world—in its context. Because of this, it's an opportunity to deal with this problem in order to handle and arrange information effectively. A Bangla phrase from a story might be something like this: an aggressive, inquisitive, essential optative, or exclamatory written document. Using the information in the data set, a number of machine learning (ML) and deep learning (DL) methods are used to classify the sentence in the text document. Our dataset is distinct because it was manually assembled while maintaining consideration for the origins and sentence structure of Bangla. The dataset is totally occupied by the raw linguistic ordinals of 16053 sentences, which was the pre-processed and severely cleaned from various unnecessary inputs. We used a supervised Deep Learning model that includes the following: Bidirectional Encoder Representations from Transformers (BERT) AND Long Short Term Memory (LSTM), Bidirectional Long Short-Term Memory (BiLSTM), Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU). Collectively, these techniques produced the final summary. The accuracy, recall, and F-measures of the suggested model were assessed in comparison to several human-made summaries. Furthermore, this study's limitations and potential implications have been explored.

Keywords— Convolutional Neural Network(CNN), Gated Recurrent Unit(GRU), Long Short-Term Memory(LSTM), Bidirectional Long Short-Term Memory(Bi-LSTM), Deep Learning.

I. INTRODUCTION

One of the most extensively spoken languages in the world is Bengali. Over 98% of Bangladeshis plus a sizable percentage of Indians speak this language as their mother's tongue. A lot of individuals prefer digital paper as it can be viewed on laptops, mobile phones, and other gadgets used in the ever-expanding realm of social media. Because of this, there are numerous Bangla news websites that offer the most recent information on a variety of subjects. A number of news based websites are also built on various online platforms like Facebook and Instagram. Every online Bangla news provider has a different layout,

style, and categorization system to convey the news items. A lot of people who read newspapers enjoy reading and evaluating news from different sources. Online news websites include subject categories and subcategories, which vary significantly across newspapers. These might not be enough to pique readers' curiosity as a result. Because readers like to research news from several sources instead of just one, providing them with relevant news depending on its contents may improve their experience. The article's main objective is to help Bengali internet news readers find relevant material by using multi-category classification. Our main objective is to construct a computerized framework for the vital task of classifying Bangla reports based on their substance, which is important to readers. Over the past few decades, many methods from machine learning and statistics have been applied to extract meaningful features and efficiently classify textual data. These supervised learning techniques require a lot of textual data to be trained. There are many large datasets available in many other languages, but only a small number of procured datasets have been developed

II. RELATED WORK

Using a large dataset, the "Bengali Word Assimilation Integrating Deep Learning Methods" project can be helpful in the following some ways. It can aid in more accurate language translation by computers. Translations may therefore seem more natural if they are made from Bengali to another language or the other way around.[3] Enhancing Virtual Assistants or Chatbots: The project's outputs may improve the intelligence and ability of Bengali chatbots, or virtual assistants, to comprehend and react in a more human-like manner. Apps for Learning Languages: This initiative could improve the efficacy of language learning applications if you're attempting to learn Bengali. It might demonstrate how well-fitting foreign words are into Bengali sentences.

A. Literature Review

The majority of research on the English language has been conducted because it is an international language. However, study on the language of Bangla language also exists. A proposal for an agrarian a brief from a passage was put forth by Kamal Sarkar [11]. After building the model in three main stages, he was able to use the model to obtain an agrarian summary in an article that was generated by a machine. By creating a distance circular augmented paragraph chart from a document, Jingzhou

Liu [12] tackled a graph-based document-based unsupervised extraction technique that allows the model to perform more fine-grained modelling of sentences and more accurately characterise the document's initial structures. Regretfully, despite the novelty of their approach, their summary generation was appalling. Many studies on the task of text classification using English along with other non-Indian indigenous languages have been conducted. A few uses for text categorization are spam detection, document classification, named entity recognition, and identifying products or contents in platforms like news businesses, blog posts, and e-commerce. Regional Indian languages are aesthetically rich, less resource-rich, and agglutinative by nature [13]. Therefore, in comparison to English natural language, developing intelligent systems for real-time applications is quite challenging. Research on automatic news classification is very popular. Nonetheless, there are a lot less studies done as it involves categorising Bangla news. Furthermore, it is rare to find a classification method for the Bangla language that solely relies on machine learning. Using the Support Vector Machine, on a data set of 150 thousand tagged news items, Umid et al. [18] accomplished a fantastic job of classifying online news. Furthermore, they classified their data using K means. Abu et al.'s method [19] for classifying Bangla news via the Nave Bayesian text classifier was proposed. With Bangla dictionary stemming, they used a web scrapper to gather additional online news texts from websites. With an accuracy of 92.79 percent, Quazi et al. [20] presented an enhanced classification of texts strategy in 2018. They used TF-IDF in conjunction with a TF threshold reached to extract features.

III. METHODOLOGY

The work methodology involves several crucial steps. It commences with manual collection of the dataset from social media platforms like Facebook. Subsequently, data preprocessing takes precedence, recognizing its pivotal role in preparing data for effective deep learning models. Tokenization follows, splitting the dataset into training and testing subsets. The subsequent phase involves model creation, drawing upon five distinct deep learning models: CNN, GRU, BiLSTM, LSTM, and BERT.

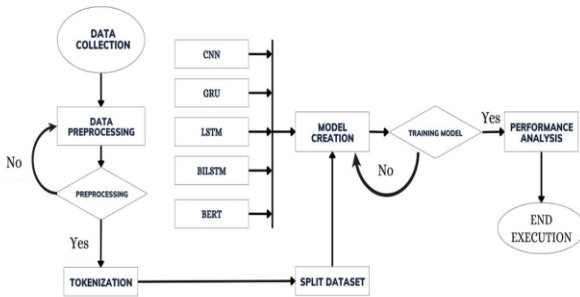


Fig. 1. The Methodology Diagram.

A. Data Preprocessing

First, we performed a statistical examine of the Bangla text in order to prepare our unbalanced dataset of Bangla News articles over grouping in this section (data acquiring it, data cleaning, and data analysis). Next, we use TF-IDF vectorizer in conjunction with new feature text data to convert the information to sequence in order to extract features. Dividing the dataset into half, and then using a sampling technique on the training dataset to balance. In the end, we used every classifier we had and determined which performed the best when it came to performance analysis. Finally, we use LIME and SHAP, two explainable AI techniques, to interpret the top model. The process of converting unclear information into a dataset with no contaminants is referred to as cleaning data. Multiple document filters were used in this experiment, along with tokenization, punctuation removal, stop words, and redundant special characters, all of which assisted in number sorting of the categories [17]. The process of breaking up a text document into smaller pieces known as tokens is known as the process of tokenization. Furthermore, tokens may be sub-words, individuals, or phrases. Space was used in the present study to keep all worlds apart from the content. Unwarranted particular characters, symbols, numbers, etc. are present in raw text data[14]. Here, the superfluous items have been eliminated using regular expressions. In order to make the data more readable, extraneous symbols and punctuation, such as [., "''—'()!], have been eliminated from the dataset. Every symbol was replaced with a space in order to break down the claim into its component words. In any written data, words with a high frequency are prevalent. They are present in every content category, but they have no such impact on the content itself. These words are referred to as "stop words." These are a few instances of stop words: " ", " ", " ", " ", etc. To obtain more precise models, all those words that stop have been removed from the text. Then the data was differentiated in the labels as religious, threat, troll, hate, vulgar, insult. As shown in the label columns below:

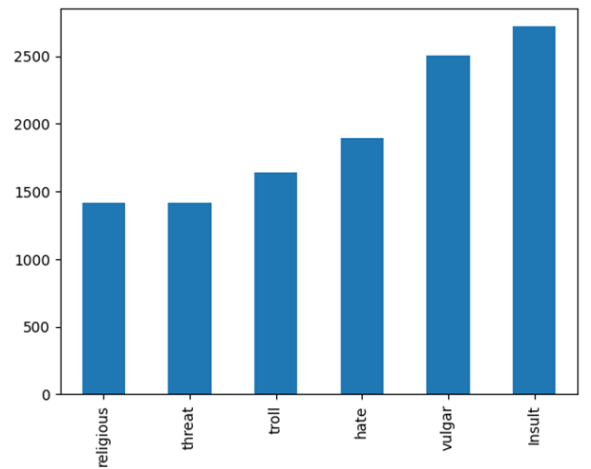


Fig. 2. Data Labeling Diagram

As we also measured the sentence Length in order to understand the datasets caliber better, a diagram of the sentence length distribution count diagram is shown below.

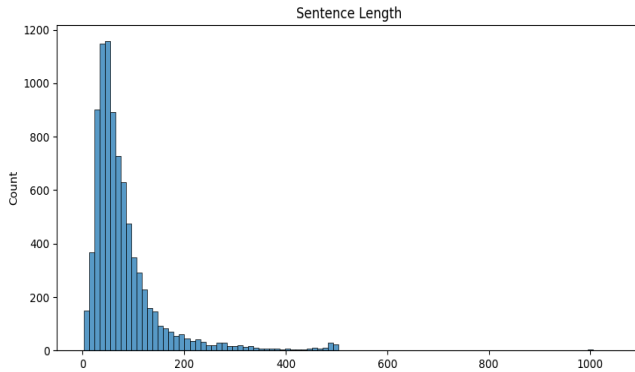


Fig. 3. Sentence length count diagram

IV. IMPLEMENTATION

This chapter presents the purpose of project our approach to the research the environment setup, the data collection methods, the material used, as well as the manner in which it was used. It defines the target group or subjects of the research and assesses the reliability of the finding.

A. Invironment Setup

To finish the project, we used Python 3.8 as the programming language and the Anaconda distribution. The Python library contains a number of machine learning libraries. Python libraries such as NumPy, Seaborn, Matplotlib, NumPy, and Scikit-learn have been used. Pandas was utilized for data manipulation and analysis. Based on Python, Pandas is incredibly quick, adaptable, and easy to use. On the other hand, a wide range of mathematical functions with high-level syntax for working with massive, multifaceted arrays and matrices are supported by the NumPy Python library. A machine learning library for the programming language Python is called Scikit-learn. We made use of Seaborn and Matplotlib for data visualization.

B. Model Implementation

In this part we are going to give a brief explaining of the algorithms we used for our categorization procedure part by part.

i) Convolutional Neural Network (CNN):

The code sequence outlines a comprehensive text classification process using a Convolutional Neural Network (CNN) within the Keras framework in Python. It encompasses several crucial stages, commencing with data preprocessing. A 'Tokenizer' object initializes the tokenization process, specifying a maximum vocabulary of 10,000 words ('max words') and an out of vocabulary token for unseen terms. This tokenizer then learns the vocabulary from the training text data ('texts train') and transforms the text sequences into integer sequences for both the training and test sets ('sequences train' and

'sequences test'). These sequences are subsequently standardized to a maximum length of 100 ('max len') using padding or truncation through the 'pad sequences' function, ensuring uniform input size for the neural network. Moving to model construction, the architecture involves layers tailored for text analysis. An embedding layer maps tokenized integers to dense vectors of 50 dimensions ('embedding dim = 50'). A 1D convolutional layer ('Conv1D') with 64 filters and a size of 3, employing Rectified Linear Unit (ReLU) activation, captures local patterns within the sequences. The subsequent global max pooling layer ('GlobalMaxPooling1D') condenses the information for effective feature extraction. The model concludes with a dense layer ('Dense') utilizing a softmax activation function, facilitating multi-class classification by predicting class probabilities. The model is compiled using the Adam optimizer with a learning rate of 0.001, employing categorical cross entropy as the loss function and accuracy as the metric for evaluation. Training occurs over 10 epochs with a batch size of 32, utilizing 20% of the training data for validation to monitor the model's performance. Post-training, the model is evaluated on the test data, computing loss and accuracy. Predictions are generated for the test set, and comprehensive classification metrics (precision, recall, F1-score) are computed along with a confusion matrix to showcase the model's performance across various classes. Finally, a heatmap visualization of the confusion matrix is plotted using 'matplotlib' and 'seaborn', illustrating the predicted versus actual classes to provide an intuitive understanding of the model's predictive capabilities across different categories. Overall, this code encapsulates a robust text classification workflow, encompassing preprocessing, model development, training, evaluation, and insightful result visualization

ii) Gated Recurrent Unit (GRU):

This script encompasses a comprehensive text classification workflow using a GRU-based neural network within Python's Keras and TensorFlow frameworks, accompanied by essential data preprocessing and result analysis steps. The initial phase involves importing necessary libraries such as NumPy, Pandas, scikit learn, TensorFlow, and Matplotlib. The dataset is assumed to be structured with text data under the 'cleaned' column and corresponding categorical labels in the 'tag' column. The data preprocessing pipeline begins by tokenizing the text using a 'Tokenizer' object, considering a vocabulary size of 10,000 words and an out of vocabulary token. Texts are converted into sequences of integers and padded/truncated to a fixed length of 50 tokens, preparing them for model ingestion. Label encoding is employed to transform categorical labels into numerical format for model training. The dataset is split into training and testing subsets using an 80-20 split ratio. A learning rate scheduler function ('lr scheduler') is defined to adjust the learning rate dynamically during model training, reducing it by half every three epochs to potentially improve convergence. The neural network architecture consists of an embedding layer mapping the

tokenized integers into dense vectors of dimension 64. Two stacked GRU layers with varying units and dropout rates are employed for sequential data processing, by fully connected layers with ReLU and softmax activations, respectively. The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy as the loss function. It's then trained on the prepared training dataset for 20 epochs with a batch size of 32, utilizing a 20% validation split and employing the defined learning rate scheduler as a callback. Following model training, evaluation on the test set is conducted, computing the loss and accuracy. Predictions are generated on the test set, and a confusion matrix is constructed to visualize the model's performance across different classes. Additionally, a classification report is generated, presenting comprehensive metrics like precision, recall, and F1-score for each class. A heatmap visualization of the confusion matrix is plotted using 'seaborn' and 'matplotlib', aiding in the intuitive interpretation of the model's predictive performance across various class predictions and actual labels. This script encapsulates a complete workflow from data preprocessing, model construction, training, evaluation, and result visualization for text classification tasks followed

iii) *BiLSTM (Bidirectional LSTM):*

Building a text classification model using a Bidirectional LSTM neural network within the context of Bengali language text data. It starts by loading necessary libraries, importing datasets, and conducting initial data preprocessing steps. The code meticulously prepares the dataset, ensuring it contains sufficient text length and doesn't have missing values, creating a clean foundation for model building. The heart of the code lies in constructing the neural network architecture. It employs the Keras Sequential API to assemble the model, beginning with an Embedding layer for converting words into dense vectors. Following this, a Bidirectional LSTM layer is employed, known for its ability to capture context from both past and future tokens. Notably, an additional Dense layer is included, incorporating L2 regularization (with a specified regularization rate) to prevent overfitting by penalizing large weights. This Dense layer is activated using ReLU, aiding in introducing non-linearity to the model. The model is compiled using the Adam optimizer with a specific learning rate and categorical cross-entropy as the loss function, which is well-suited for multi-class classification tasks. The architecture is trained for 25 epochs using the training dataset, with parallel validation on a separate subset. During training, the code visualizes the model's learning progress by plotting the loss and accuracy curves across epochs, facilitating a deeper understanding of how the model's performance evolves over training iterations. This visual insight helps gauge the model's convergence and performance stability. The model's performance is thoroughly evaluated. The code calculates and reports accuracy metrics for both the training and testing datasets. Furthermore, it generates a confusion matrix, providing a visual representation of the

model's predictions against actual labels across different classes. Moreover, a comprehensive classification report is generated, offering detailed precision, recall, and F1-score metrics for each class, aiding in understanding the model's strengths and weaknesses in classifying Bengali text data.

iv) *LSTM (Long Short-Term Memory) and BERT (Bidirectional Encoder Representations from Transformers):*

Beginning with meticulous data preprocessing, including text cleaning and tokenization using the Bangla BERT tokenizer, it meticulously readies the dataset for training, validation, and testing. The model architecture intricately weaves together BERT's contextual understanding with Conv1D and MaxPooling1D layers, culminating in a Bidirectional LSTM layer fortified with attention mechanisms to refine sequence comprehension. The output flows through dense layers with sigmoid activation for multi label classification. Post compilation with binary cross entropy loss and Adam optimizer, the model undergoes rigorous training and validation, meticulously tracking its performance across epochs. Subsequently, it undergoes evaluation on a separate test dataset, gauging metrics like accuracy and loss. Additionally, the code extends its analysis to plotting ROC-AUC curves, assessing the model's discrimination across various classes, and employs LIME explanations to offer insights into the model's decision making process, elucidating its predictive behavior on specific text instances. This comprehensive approach amalgamates cutting edge techniques, presenting a robust solution for complex multilabel text classification tasks in Bengali, backed by thorough evaluation and interpretability measures for deeper insights into the model's functionality Result Analysis

V. RESULT ANALYSIS

The performance analysis of these machine learning algorithms, based on metrics such as precision, recall, F1 score, and accuracy, provides valuable insights into their capabilities for classification tasks. The Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) showcase relatively balanced performances. CNN demonstrates consistent and even performance across the metrics, maintaining precision, recall, F1 score, and accuracy around the 70- 71% range. Similarly, GRU maintains reasonable accuracy, albeit with slightly lower precision, recall, and F1 score averaging around 66-67%. In contrast, the Bidirectional LSTM (BiLSTM) presents an intriguing trade-off. It achieves a higher recall rate of around 65% but at the expense of precision, which is notably lower at approximately 58%. This indicates BiLSTM's proficiency in capturing a larger number of true positive instances while also generating more false positives compared to the other models. The standout performer among these algorithms is the LSTM model augmented with BERT. It attains an impressive precision of 81%, a recall of 76%, and an F1 score of 78%, showcasing a superior ability to accurately identify positive instances while maintaining a balanced overall performance shown in Figure 4.

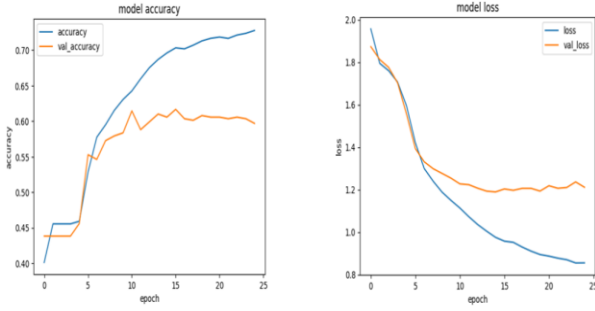


Fig. 4: Model Accuracy and Loss

Firstly, the micro-average ROC curve, boasting an area of 0.94, consolidates performance by considering aggregated true positive and false positive rates across all classes. Secondly, the macro-average ROC curve, with an area of 0.93, provides a balanced evaluation across individual class ROC curves. Moreover, individual ROC curves for each class demonstrate discriminative ability, displaying areas ranging from 0.90 to 0.95. These specific class ROC curves indicate how well the model distinguishes each class from the rest.

Table 1: Performance Metrics of Different Models

Algorithm	Precision	Recall	F1 Score	Accuracy
LSTM+BERT	81%	76%	78%	75%
CNN	70%	71%	70%	71%
GRU	67%	66%	63%	66%
BiLSTM	58%	65%	62%	65%

A. Comparative Result Analysis

Here we represent a Table 2 with comparing with other research works deliberately done by researchers. "Haque [24]" exhibited 58% precision, 51% recall, 48% F1 score, and 61% accuracy. "Mahimul Islam [25]" demonstrated 48% precision, 52% recall, 62% F1 score, and 59% accuracy. "Kamal Sarkar [26]" showcased 62% precision, 57% recall, 62% F1 score, and 61% accuracy. Notably, the "LSTM+BERT" approach achieved substantially higher metrics across the board, boasting 81% precision, 76% recall, 78% F1 score, and 75% accuracy. This comparison highlights the superior performance of the "LSTM+BERT" model in this specific task compared to the other referenced methodologies or studies.

Table 2: Comparative Result Analysis

Other Works	Precision	Recall	F1 Score	Accuracy
Haque [24]	58%	51%	48%	61%
Mahimul Islam [25]	48%	52%	62%	59%
Kamal Sarkar [26]	62%	57%	62%	61%
LSTM+BERT	81%	76%	78%	75%

VI. CONCLUSIONS

In the current era, there is a need for an efficient way to categorize Bangla text, and the primary objective of this research project is to create a model for doing just that. And by leveraging deep learning technologies, we were able to accomplish this with exceptional accuracy. The identical text data are divided into various categories in this dataset. Models were incorrectly classified and unable to learn successfully as a result. The suggested hybrid

model outperformed other single deep neural network models in terms of performance, earning an accuracy of 75% across seven categories. The categorization of texts is a very difficult task because there aren't enough trustworthy sources or standard, balanced datasets available in the Bangla language. If there are sufficient Bangla language resources available, these models for deep learning can yield better results than this one. Future research may therefore concentrate on producing a uniform balanced Bangla dataset in addition to creating a more effective model for this NLP text categorization issue.

A. Future Recommendation

Supervised knowledge is utilized to achieve Bengali document categorization in this instance. Therefore, there remain more research experiments that need to be conducted without supervision. Other than the currently published works, such as cosine similarity measures, can be investigated through the use of similarity metrics. The size of the dataset should be expanded in the future to create a sophisticated and uniform output measurement system. Coherent, intricate, and larger datasets will require analysis using improved kernels that effectively perform the classification tasks.

B. Limitation

The main limitation we faced was the limited resources. As Bengali NLP research is an ongoing paradigm of the research contradiction, we are hoping for the better resources will soon come in order for the improvement of the technological evolution.

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